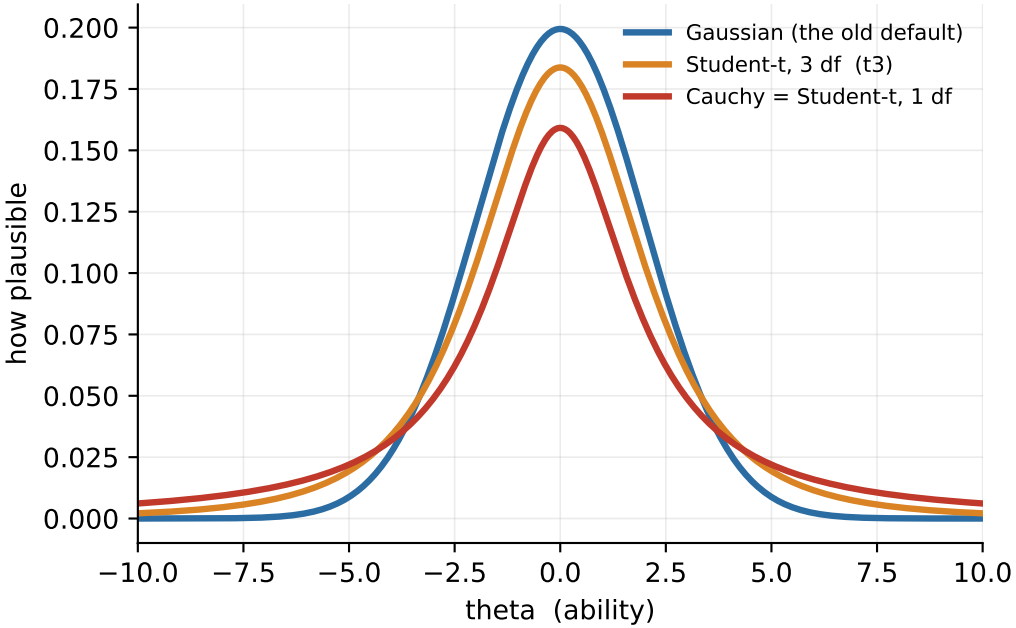
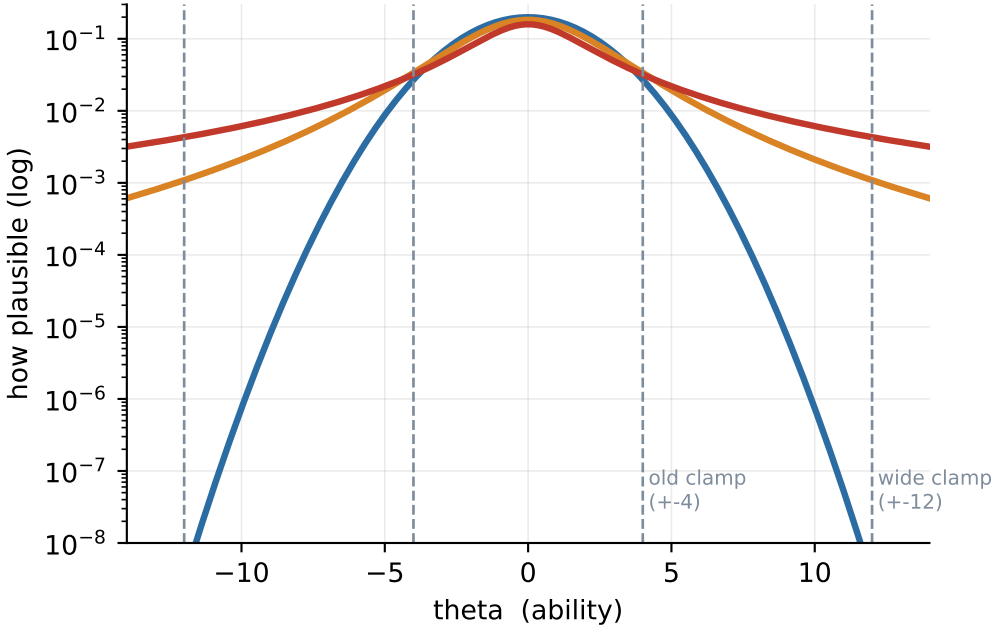
What is an ability prior?

Every subject gets one number, theta, for how good it is (higher = better). Before we see any of its answers, we still have to guess a value. The prior is that guess, drawn as a hill: the higher the hill at a value, the more plausible that value is BEFORE seeing any labels.

A. The three hills (same width, different tails)



B. Same picture, log scale: zoom into the tails



How to read it

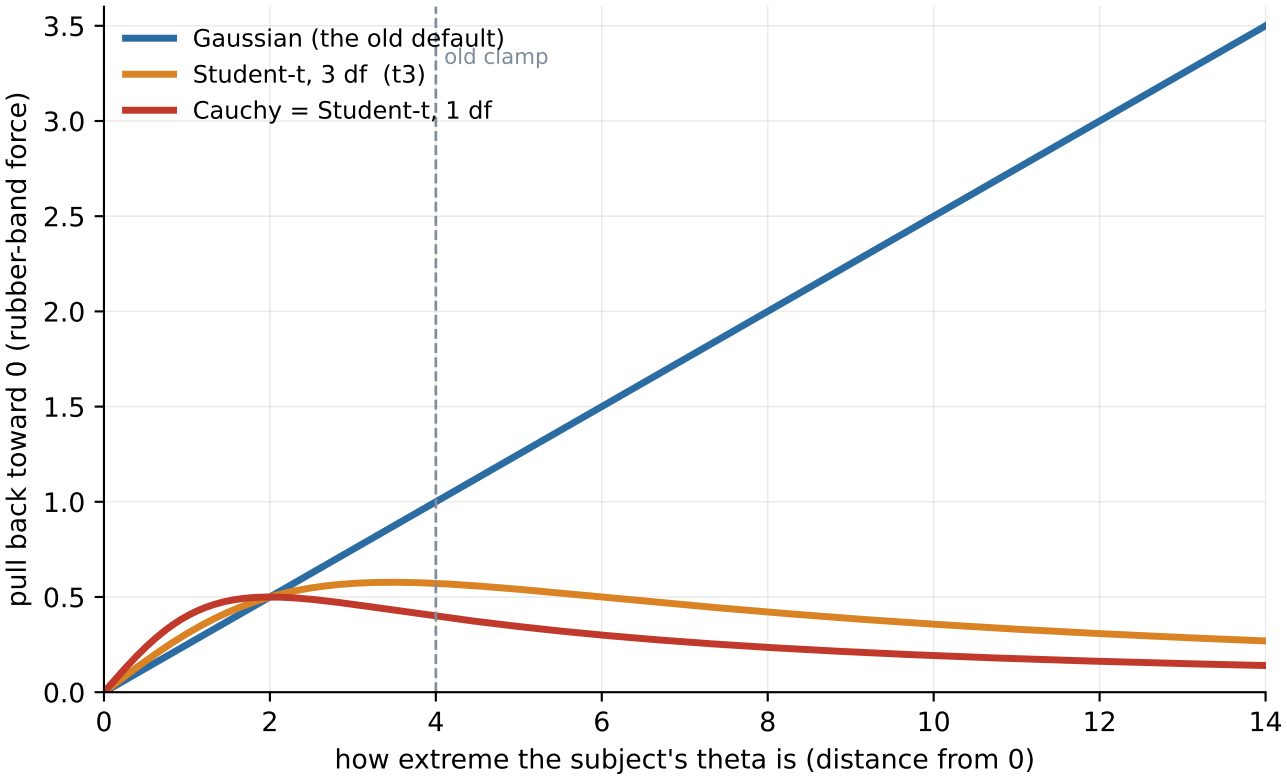
- Near the middle all three look alike. They only disagree about EXTREME subjects.
- Gaussian: extreme values are very unlikely. At theta = 8 it is about 3,000 times less plausible than at 0.
- Student-t / Cauchy: extreme values are rare but possible. They have 'heavy tails'.
- Cauchy has the heaviest tails, t3 is in between, Gaussian has the lightest.
- The dashed lines are the CLAMP: a hard wall. The solver is not allowed to put theta beyond it, no matter what the answers say. 'Old clamp' = +-4, 'wide clamp' = +-12.

Why we care

Some subjects get almost everything wrong (one deepseek run scored 0.8%). Their true theta is very negative.
If the prior says 'that is nearly impossible', it keeps pulling them back toward the middle, and we predict they will do better than they do.

How hard does each prior pull?

The prior acts like a rubber band tied to zero. The further the subject's theta is from zero, the more the band pulls it back.
The question is how the pull changes as theta gets more extreme.



What the curves say

Gaussian (blue)
The pull keeps growing the further out you go. At theta = 12 it is tugging 3 units hard. Evidence has to shout louder and louder to move it.

Student-t (orange, red)
The pull rises, peaks, then FADES. Far out, the band goes slack and the answers decide. Cauchy fades the fastest.

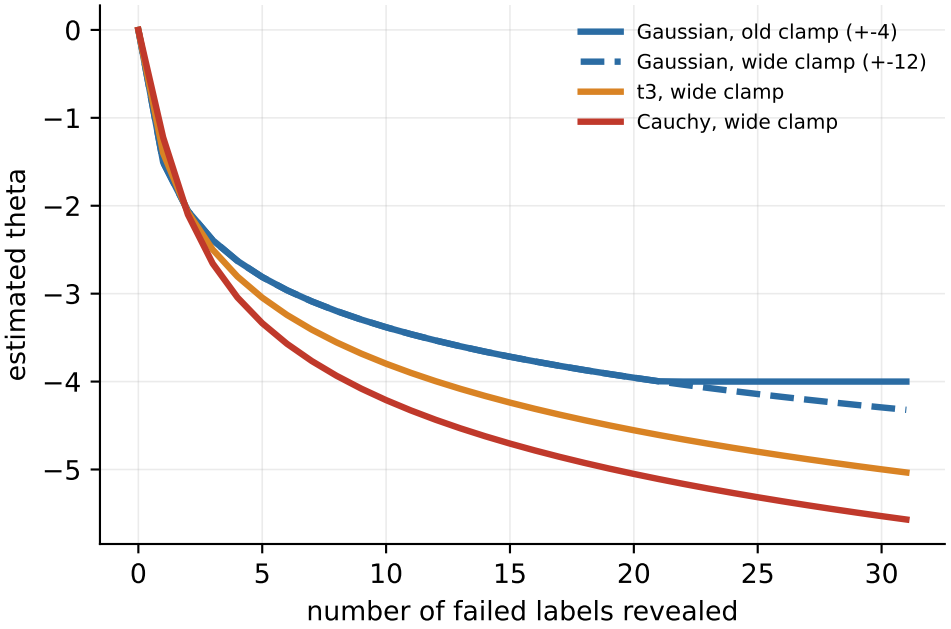
In plain words

- Ordinary subjects (theta near 0): all three priors pull about the same. No difference.
- Extreme subjects: the Gaussian keeps dragging them back. The heavy-tailed priors let go.
- That is the whole idea: shrink small, uncertain estimates, but do not crush a subject whose labels clearly say 'always fails'.
- Trade-off: letting go also means a few unlucky labels can push a normal subject too far.
  That is why we also tried the middle option (t3) and not only Cauchy.

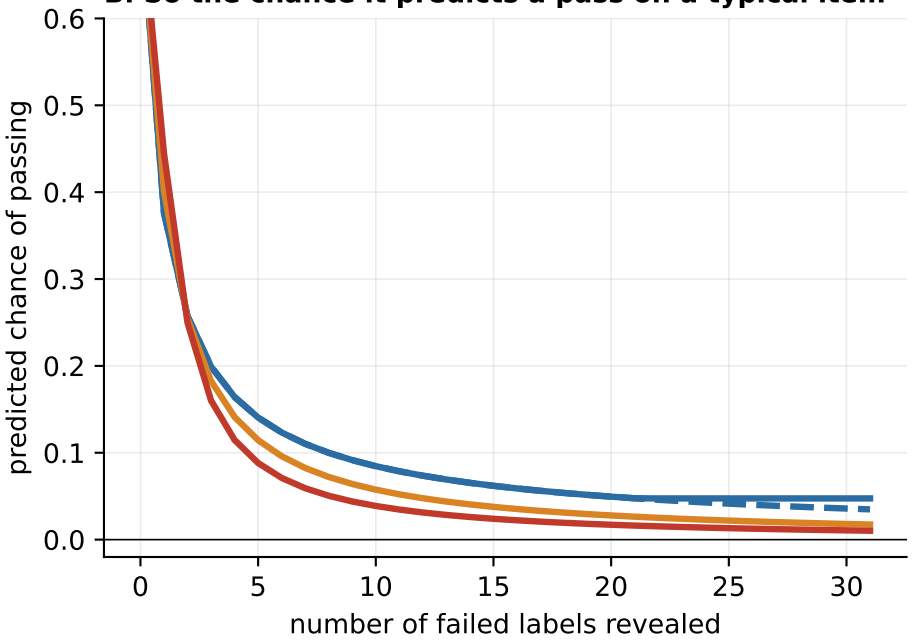
A worked example: a subject that fails everything

We reveal labels one by one. Every label is a FAIL, on items that most subjects pass (easy items).
The plots below use the real solver from our model, so this is what the code would do, not a cartoon.

A. What the solver believes theta is



B. So the chance it predicts a pass on a typical item



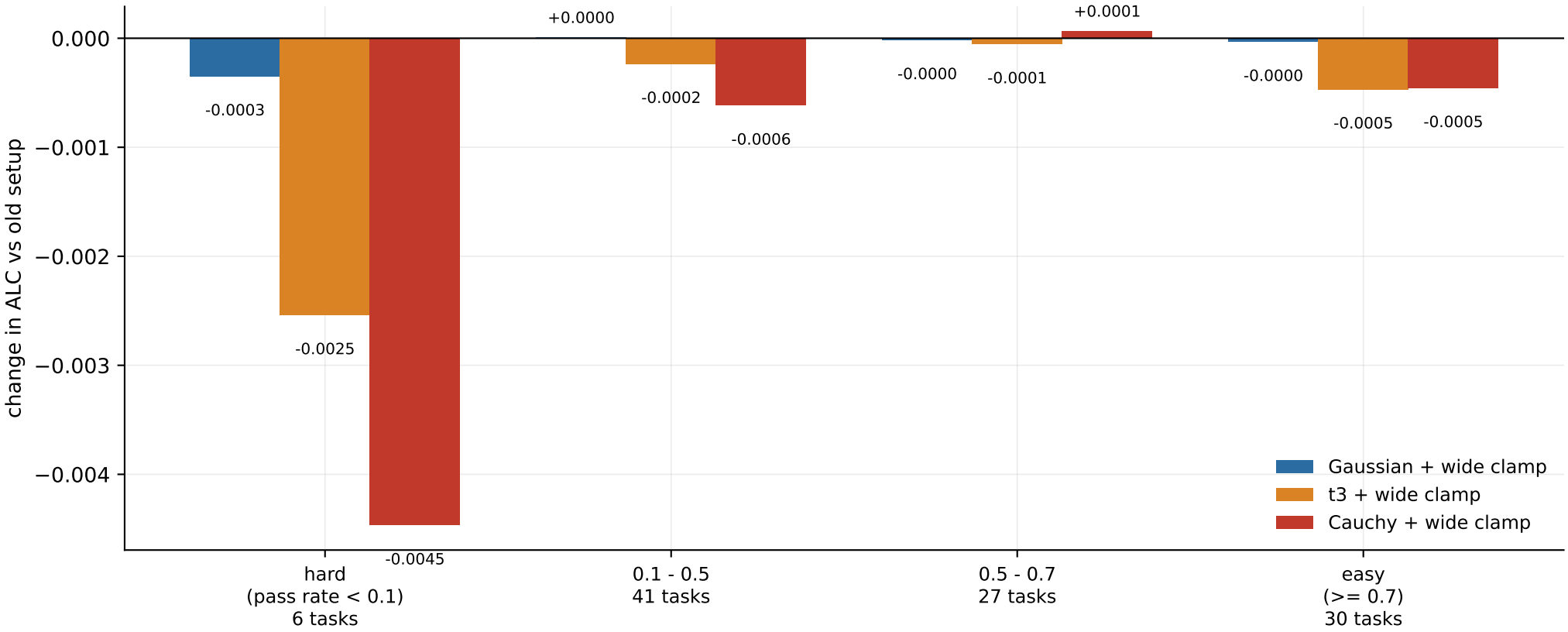
What to notice

- After 31 straight failures: Gaussian with the old clamp says theta = -4.0, Gaussian with the wide clamp -4.3, t3 says -5.0, Cauchy says -5.6.
- The flat part of the solid blue line at -4 is the CLAMP (a wall), not the prior.
- Remove the wall (dashed blue) and the Gaussian still slows down a lot: its own pull keeps growing, so it only creeps lower. Widening the clamp alone changes little. That matches our result.
- The heavy-tailed priors keep sliding down as the failures pile up, so the predicted pass chance keeps shrinking, as it should.
- At 0 labels all four are identical: with no evidence, the prior is the guess.

Setup: items of difficulty -1 (an average subject passes about 73%), one ability dimension, prior width 2 (variance 4).

Did it help? Measured results

Change in Brier ALC against the old setup (Gaussian, +-4 clamp). Lower is better, so BARS GOING DOWN = BETTER.
Tasks are grouped by how often the subject passes. Model: me5 embeddings + 4-dimensional ability. Validation set, one seed.



What it says

- Widening the clamp alone (blue) does almost nothing. The prior is the problem, not the wall.
- Heavy tails help the hard tasks a lot: Cauchy improves them by about 0.0045, t3 by about 0.0025.
- The easy tasks are not hurt after calibration (all within 0.0005 of the old setup).
- Overall the gain is small, about 0.0004 to 0.0006, because only 6 of 104 tasks are hard.

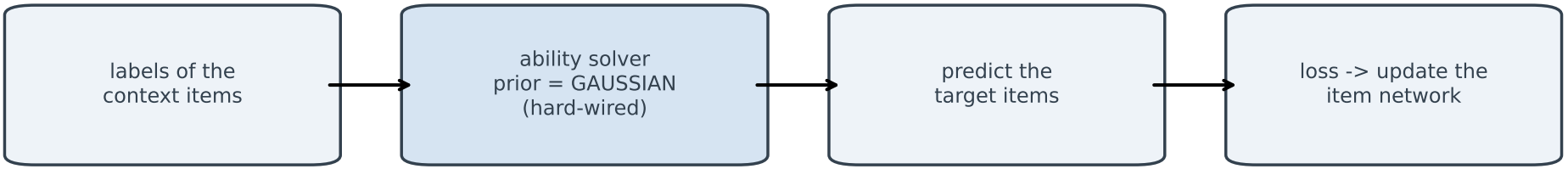
Be careful

- One seed and only 6 hard tasks: a lead, not a proven result.
- Before calibration, Cauchy made the easy tasks worse (+0.0038). It looks fine only after calibration, so t3 is the safer pick.

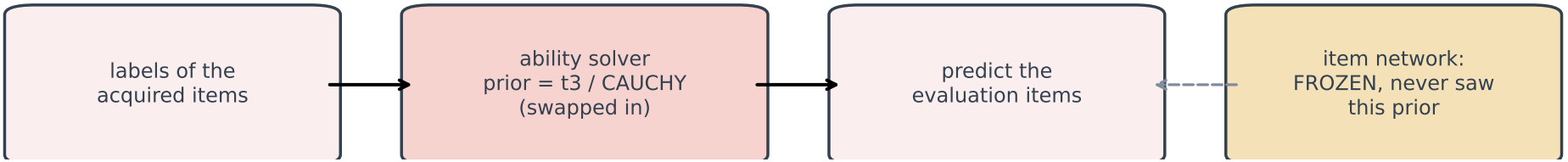
The mismatch: one prior in training, another at test time

Our model runs the same ability solver (Newton MAP) in two places. Only one of them got the new prior.

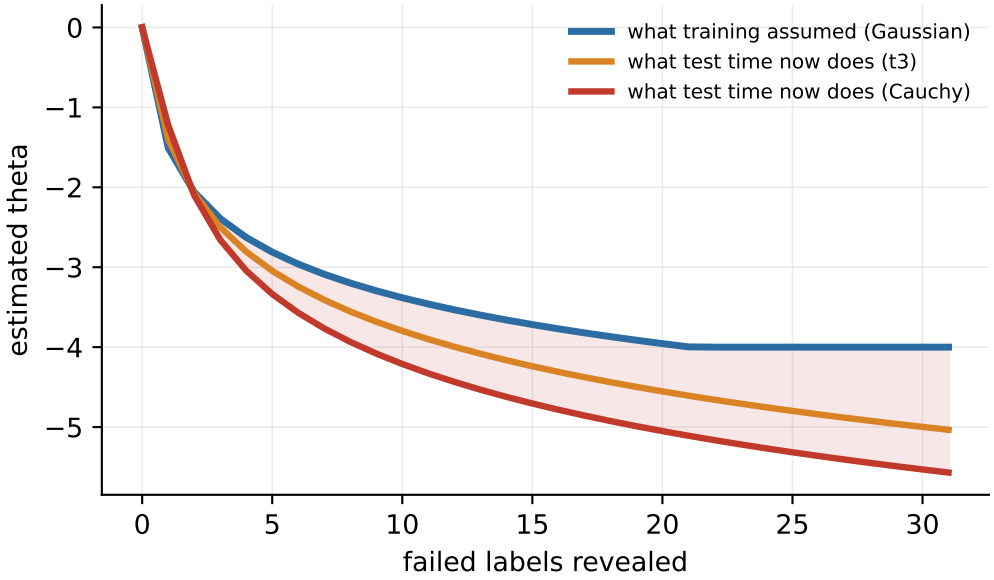
TRAINING (item network learns difficulty and discrimination)



TEST TIME (what we changed)



The shaded gap is the mismatch



Why it matters

- The item network was trained while the Gaussian kept shrinking theta. It may have learned difficulties that quietly make up for that shrinkage.
- At test time we remove the shrinkage, so those difficulties are slightly off.
- Even so, heavy tails helped the hard tasks (one seed, 6 tasks). Retraining with the new prior MIGHT help more, but we have not tested it.

Not done yet: retraining with the t prior in the training loop (needs a GPU run).